

Real-time Short-term Live Fuel Moisture Estimation through Hybrid AI-Physics Modeling for Enhanced Wildfire Prediction

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1. Introduction and Motivation

In recent years, the frequency, intensity, and unpredictability of wildfires across the United States have surged, driven by the confluence of climate change, prolonged drought, and expanding human development at the wildland-urban interface (WUI). This growing threat has exposed critical limitations in our capacity to anticipate fire behavior at spatial and temporal scales relevant to operational decision-making. Among the most influential factors in wildfire ignition, spread, and intensity is **Live Fuel Moisture Content (LFMC)**—the water content of living vegetation—which directly governs fuel flammability. However, despite its importance, LFMC remains notoriously difficult to monitor and forecast in real time, particularly at fine spatial resolutions needed for tactical fire management (Yebra et al., 2013). Current operational LFMC products, such as those available through the NCAR Fuel Moisture Dashboard, primarily rely on coarse-resolution optical remote sensing (e.g., VIIRS at ~375m) and offer limited forecasting capability (Schreck et al., 2023; NCAR Dashboard, 2024). These systems provide important regional context but fall short in capturing the spatial heterogeneity and short-term dynamics of vegetation moisture—especially in ecologically and operationally sensitive areas such as chaparral-dominated landscapes or densely populated WUIs. Emerging research (e.g., Rao et al., 2020) has demonstrated that **synthetic aperture radar (SAR)** from Sentinel-1, when paired with machine learning (ML) methods, can retrieve LFMC at finer spatial resolutions (~10m, Rao et al., 2020). SAR's cloud-penetrating capabilities and consistent temporal revisit make it well-suited for continuous monitoring. However, such approaches often remain confined to nowcasting, lack physical interpretability, and are rarely validated with robust in-situ measurements. Moreover, they typically overlook the potential of integrating **numerical weather prediction (NWP)** and **AI-based forecasting** to extend predictions beyond current observational bounds (Pathak et al., 2023).

This project responds to these limitations by developing a **next-generation LFMC forecasting system** that fuses physics-informed machine learning, SAR and optical remote sensing, environmental proxy modeling, and ecohydrological realism. Our system aims to generate **daily LFMC maps at 10-meter resolution with up to 10-day lead times**, providing the spatial granularity, temporal foresight, and physical transparency required for effective fire management in a warming world. We build upon our promising preliminary results in which LightGBM models (Ke et al 2017) with extensive hyperparameter tuning trained on Sentinel-1 SAR achieved improved LFMC estimation accuracy (RMSE ~17.7% from 25%) over previous efforts. This serves as a foundational proof of concept. Our goal now is to operationalize and scale this capability, while embedding physical realism and field-based validation to enhance generalization, trust, and usability for agencies such as CAL FIRE, USFS, and wildfire forecasting centers. Our proposed system represents a major advancement over existing tools by: (i) **Achieving high spatial resolution (10m)** to support local-scale fire behavior modeling; (ii) **Integrating forecasting inputs (3–10 day)** from both physics-based NWP and AI-based systems; (iii) **Incorporating ecohydrological processes and constraints** to improve interpretability and performance under

extreme climatic conditions; (iv) **Validating outputs through destructive LFMC sampling** across ecoregions; and (v) **Delivering a modular, dashboard-ready platform** tailored to the real-time needs of wildfire managers.

Through this work, we aim to create a scalable, interpretable, and field-validated LFMC forecasting system that meets the urgent demand for high-fidelity, predictive fire risk intelligence—transforming LFMC from a lagging indicator to a leading variable in fire management strategies.

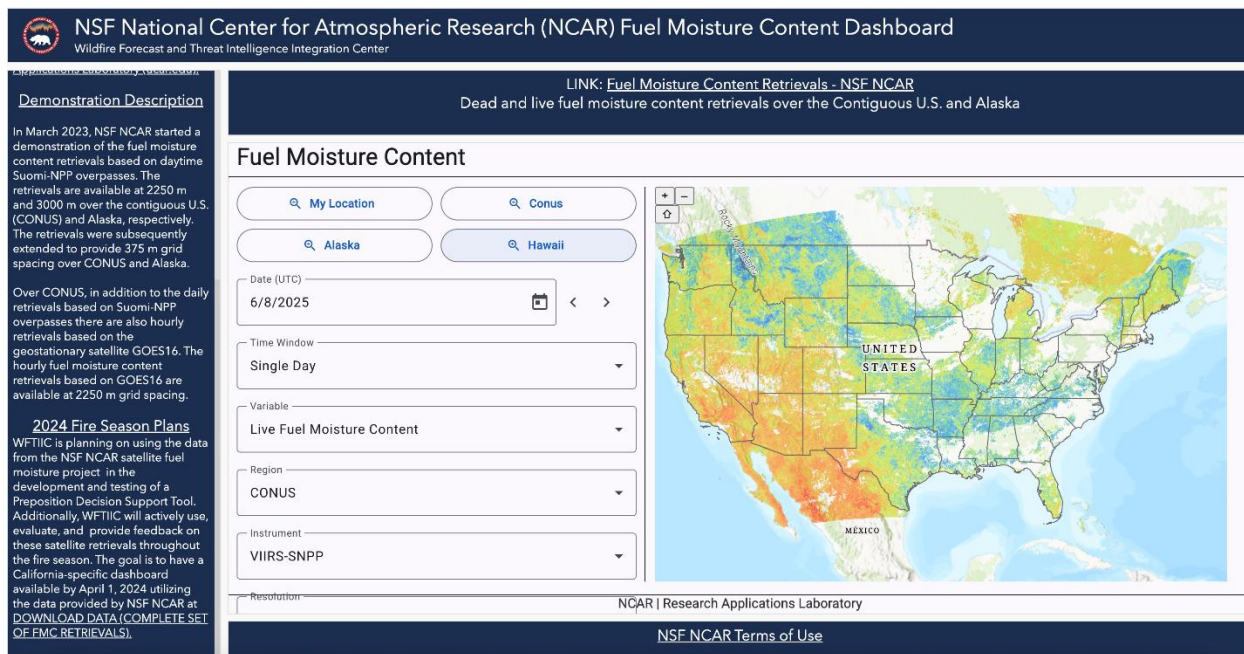


Figure 1: NSF National Center for Atmospheric Research (NCAR) Fuel Moisture Content Dashboard. The goal of this project is to develop a dashboard of live fuel moisture content forecasts up to 10 days lead time

2. Objectives and Research Questions

Goal

The primary goal of this project is to develop a high-resolution, physically informed, AI-based system to nowcast and forecast Live Fuel Moisture Content (LFMC) across the continental U.S. (CONUS). The system will prioritize interpretability, generalization, and field-validated accuracy, while operating at a 10-meter spatial resolution, matching the capabilities of Sentinel-1 synthetic aperture radar (SAR). This represents a major advancement over existing operational products—such as the VIIRS-based NCAR LFMC dashboard—which are limited to much coarser resolutions (typically 375–500 m), restricting their utility for local-scale fire behavior forecasting and tactical fire management. LFMC is a critical variable influencing wildfire ignition, spread, and intensity, yet it remains difficult to monitor and predict at spatial and temporal scales relevant to operational fire agencies (Yebra et al., 2013, Peterson et al., 2008). Prior studies have shown that SAR, particularly from Sentinel-1, can retrieve LFMC (Rao et al., 2020), but challenges remain in forecasting, physical interpretability, and ground validation.

This project will integrate SAR, optical remote sensing, numerical weather prediction (NWP), AI-NWP ecohydrological modeling, and machine learning to deliver a scalable, interpretable, and field-validated LFMC forecasting system with 3–10 day lead times. The high spatial resolution will enable actionable insights for wildfire risk mitigation, especially in complex, heterogeneous landscapes such as the wildland-urban interface (WUI).

Research Questions

To realize this goal, we pose the following research questions:

1. Which machine learning architectures best model LFMC spatiotemporal patterns from SAR and ancillary remote sensing data at 10-meter resolution?: We will test a suite of machine learning approaches—such as gradient boosting machines (e.g., XGBoost, LightGBM) and advanced deep learning models (e.g., FT-Transformer (Gorishniy et al 2021), TabNet, NODE)—to identify architectures capable of capturing fine-scale LFMC variability with spatial and temporal generalization.

2. What environmental variables can substitute SAR inputs to enable LFMC modeling in regions or periods where SAR data are unavailable?: Using environmental predictors (e.g., ERA5 soil moisture, VPD, radiation, temperature), we will build proxy models to estimate LFMC at Sentinel-1 resolution even in SAR-limited contexts, applying feature attribution techniques like SHAP for interpretability.

3. Can modern AI-driven weather forecasts support accurate 3–10 day LFMC forecasting at high spatial resolution?: We will integrate NWP outputs (e.g., from GFS, IFS) and AI-based models (e.g., GraphCast, FourCastNet) into the LFMC forecasting pipeline to evaluate their ability to provide fine-scale, forward-looking predictions, including uncertainty quantification by forecast lead time. **4. How can ecohydrological modeling and field-based validation improve the physical realism and operational trust of LFMC forecasts?:** By incorporating physically meaningful variables (e.g., plant water stress, soil moisture deficits) from ecohydrological models and conducting field-based destructive sampling of LFMC, we aim to ground ML outputs in empirical evidence and improve model calibration across vegetation types.

3. Technical Approach: Five-Stage Plan

Step 1: Synthetic Aperture Radar (SAR) + AI/ML Modeling

The first stage of this project focuses on leveraging synthetic aperture radar (SAR) from Sentinel-1 in conjunction with advanced artificial intelligence and machine learning (AI/ML) models to produce high-resolution, spatially explicit estimates of Live Fuel Moisture Content (LFMC). This work directly addresses a key limitation of existing operational tools, such as the NSF NCAR LFMC Dashboard, which currently relies on VIIRS observations at coarse spatial resolutions (~375–500 meters). While VIIRS provides valuable wide-area context, its spatial granularity is insufficient for many localized fire management applications, particularly within the heterogeneous landscapes of the wildland-urban interface (WUI). Moreover, VIIRS is constrained by optical sensing limitations, including susceptibility to cloud cover, which can delay or obscure critical observations.

Sentinel-1 SAR offers several technical advantages that make it well-suited for LFMC estimation (Rao et al., 2020). First, its C-band radar is cloud-penetrating and provides reliable observations under diverse weather conditions. Second, it delivers data at a significantly finer spatial resolution—10 meters—which is ideal for capturing the complex biophysical patterns that influence fire behavior at operationally actionable scales. Third, prior studies (e.g., Rao et al., 2020) have demonstrated that Sentinel-1 VV and VH backscatter can be effectively used to estimate LFMC through machine learning methods, though those earlier efforts were limited in terms of physical interpretability, temporal generalization, and systematic field validation.

We build on these foundations by developing a robust AI/ML modeling pipeline designed to extract LFMC estimates from Sentinel-1 backscatter data, terrain features (elevation, slope), and land cover classifications. Formally, the LFMC estimation task is posed as a supervised regression problem:

$$y_{\text{LFMC}} = f(X_{\text{SAR}}, X_{\text{terrain}}, X_{\text{landcover}}; \theta)$$

where y_{LFMC} is the predicted live fuel moisture content, $f(\cdot)$ denotes the machine learning model with parameters θ , and the inputs X include Sentinel-1 SAR backscatter (VV and VH), terrain variables, and categorical land cover information.

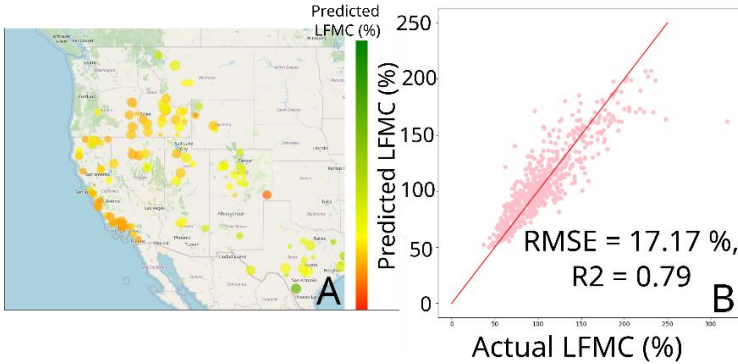


Figure 2: Preliminary results showing A- Predicted LFM (%) at different sites using Sentinel-1 SAR as input, B- Performance of hyperparameter tuned LightGBM model in estimating LFM from Sentinel-1 SAR Preliminary work conducted by our team has already yielded promising results: by using LightGBM—a state-of-the-art gradient boosting framework—we

have achieved an RMSE of 17.7% relative to the mean LFM, an improvement over previously reported benchmarks. Model accuracy is evaluated using standard statistical metrics, including root mean square error (RMSE)

In this modeling framework, we will conduct a comprehensive evaluation of both tree-based machine learning models (e.g., XGBoost (Chen et al 2016), LightGBM (Ke et al 2017), CatBoost (Hancock et al 2020)) and deep learning models optimized for tabular data, including architectures such as FT-Transformer (Gorishniy et al 2021), TabNet (Arik et al 2021), NODE (Popov et al 2019), and MLP-Mixer (Eslamian et al 2025). These models will be trained and cross-validated using gridded field-sampled LFM datasets across multiple ecoregions and seasons, to evaluate both spatial and temporal generalizability. Sentinel-1's temporal revisit interval (typically 6–12 days depending on geographic location and dual-polarization mode availability) allows for meaningful near-real-time mapping and monitoring, which is a critical capability for tactical and strategic fire planning.

The output from this stage will be a high-resolution (10-meter) gridded LFM estimation product that can be updated dynamically as new SAR data become available. This represents a substantial advancement over existing LFM maps and will serve as the foundation for subsequent modeling steps involving proxy estimation, forecasting, and physical integration. By significantly enhancing spatial resolution, maintaining interpretability, and grounding predictions in physically relevant radar signatures, this component of the project will form the cornerstone of a next-generation LFM decision-support tool designed for fire-prone ecosystems and operational end users.

Step 2: Environmental Proxy Modeling for SAR-Limited LFM Estimation

To enable continuous and spatially complete monitoring of live fuel moisture content (LFMC), this stage focuses on developing a robust modeling framework that estimates LFM using environmental variables as proxies in the absence of SAR data. While Sentinel-1 SAR enables high-resolution LFM estimation, its coverage is limited by its revisit interval (typically 6–12 days, depending on latitude and acquisition mode) and occasional data outages. Moreover, SAR observations are unavailable in certain geographic areas due to acquisition priorities, sensor viewing geometry, or historical archive gaps. To ensure full spatial-temporal operability, this step extends the system's capabilities by developing SAR-free, proxy-driven LFM models based on physically meaningful environmental predictors.

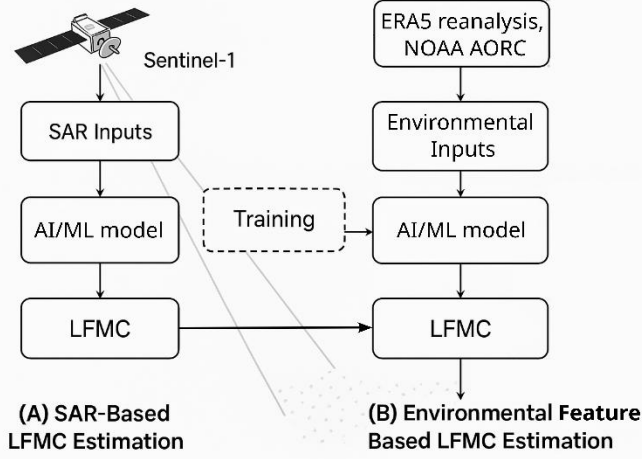


Figure 3: Proxy Modeling Framework for SAR-Limited LFM C Estimation. This figure illustrates a two-stage approach for generating high-resolution LFM C estimates. In (A), Sentinel-1 SAR inputs are processed through an AI/ML model to generate 10-meter resolution LFM C estimates. These SAR-derived LFM C outputs then serve as surrogate ground truth for training a second AI/ML model in (B), which uses environmental inputs (e.g., from ERA5 and NOAA AORC) to estimate LFM C in regions or periods lacking SAR coverage. This approach enables spatially and temporally

complete LFM C mapping by leveraging environmental features to replicate SAR-derived estimates.

We leverage multivariate input features from ERA5 reanalysis (produced by ECMWF) and the NOAA Analysis of Record for Calibration (AORC) dataset, both of which provide gap-free, regularly updated meteorological and land surface fields at fine temporal resolution. Variables include volumetric soil moisture, vapor pressure deficit (VPD), temperature (surface and root zone), shortwave and longwave radiation, precipitation, wind speed, and relative humidity. These data are used as inputs to train machine learning models that can predict LFM C at 10-meter resolution, aligning with the spatial scale of the SAR-derived products generated in Step 1. We formulate the environmental proxy modeling task as a supervised regression problem:

$$y_{\text{LFMC}} = g(X_{\text{env}}; \phi)$$

Here, y_{LFMC} is the predicted live fuel moisture content, $g(\cdot)$ is a AI/ML model (e.g., LightGBM, TabNet, FT-Transformer), ϕ denotes model parameters, and X_{env} is a vector of environmental predictors:

$$X_{\text{env}} = \{\text{SM}, \text{VPD}, \text{T}_{\text{surf}}, \text{T}_{\text{root}}, \text{RSW}, \text{RLW}, \text{RH}, \text{P}, \text{W}\}$$

To ensure consistency and realism, training targets for these proxy models are derived from the SAR-based LFM C estimates produced in Step 1. This approach not only provides a scalable training dataset but also enables seamless substitution between SAR-driven and proxy-driven estimates in operational workflows. For interpretability and feature selection, we will apply SHAP (SHapley Additive exPlanations) to quantify the marginal contribution of each predictor:

$$y_{\text{LFMC}} = \phi_0 + \sum_{j=1}^M \phi_j$$

where ϕ_0 is the expected model output and ϕ_j is the SHAP value quantifying the influence of the j -th feature. This interpretability framework supports seasonal model diagnostics, identification of dominant controls on LFM C, and regionalization strategies for generalization.

To further refine the accuracy of proxy-based LFM C estimation, we will stratify models by phenological season—spring emergence, summer peak, senescence, and dormancy—capturing vegetation phase transitions that strongly affect moisture dynamics. Additionally, we will explore ecoregion-specific modeling to account for biome-level variation in how environmental stress translates to LFM C reduction.

This methodology builds upon earlier work by Yebra et al. (2013), Yebra et al., 2018, Peterson et al., 2008, who used optical and meteorological proxies for LFM C retrievals in Australian shrublands, and recent studies such as Jones et al. (2021), which demonstrated the effectiveness

of environmental reanalysis data in capturing LFMC dynamics across the western United States. Our approach advances these studies by fusing fine-resolution training targets (from Sentinel-1), using modern AI architectures, and delivering high-resolution outputs applicable for operational fire behavior forecasting.

Expected Outcomes of Step 2: (i) Trained machine learning models that estimate LFMC at 10-meter resolution using only environmental variables. (ii) SHAP-based variable importance maps, showing dominant drivers by region and season. (iii) An operational SAR-substitute system to fill spatial and temporal gaps in LFMC coverage. (iv) Expanded utility of the overall system in persistently cloudy regions, acquisition gaps, or SAR-impaired contexts.

By enabling SAR-free estimation of LFMC, this step increases the reliability and completeness of the overall system, ensuring it can produce consistent, high-resolution LFMC maps across the continental U.S., even in observationally constrained scenarios. This capability is essential for maintaining situational awareness and supporting wildfire mitigation efforts under real-world data limitations.

Step 3: Forecast-Enabled LFMC Modeling with NWP and AI-Based Weather Inputs

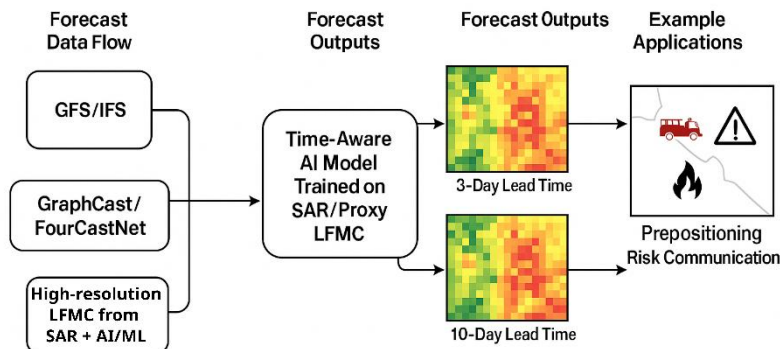


Figure 4: LFMC Forecasting Using NWP and AI-Based Weather Predictions. This figure illustrates how high-resolution forecasts of LFMC are generated by using meteorological predictions from physics-based (e.g., GFS, IFS) and AI-based (e.g., GraphCast, Lam et al., 2022, FourCastNet,

Pathak et al., 2022) weather models. These forecasts serve as input drivers to a time-aware AI model previously trained on SAR- and proxy-derived LFMC data. The output is a sequence of future LFMC maps (e.g., 3-day and 10-day lead times), enabling proactive wildfire management strategies such as firefighting resource prepositioning and public risk communication.

While Steps 1 and 2 enable retrospective and near-real-time estimation of Live Fuel Moisture Content (LFMC), operational fire management demands forward-looking forecasts to support prepositioning, resource allocation, early warning, and public communication. In this third stage, we extend our LFMC prediction system by incorporating forecast-based inputs to generate 3–10 day lead time LFMC predictions at 10-meter spatial resolution. This represents a major leap beyond existing systems, such as the NCAR VIIRS-based dashboard, which currently emphasizes near-real-time monitoring with limited temporal forecasting ability and coarse spatial granularity. We propose a novel fusion of physics-based numerical weather prediction (NWP) models (e.g., GFS, IFS) and AI-driven weather forecasting systems (e.g., Google’s GraphCast and NVIDIA’s FourCastNet) as dynamic drivers of LFMC evolution. These systems offer complementary benefits: traditional NWP provides physically grounded forecasts with mature uncertainty estimates, while AI-NWP models offer finer spatiotemporal resolution, better representation of extreme events, and dramatically lower inference costs. Together, these provide a robust foundation for next-generation LFMC forecasting. We define the LFMC forecasting task as a supervised spatiotemporal prediction problem, where environmental inputs from forecast models are mapped to future LFMC values:

$$y_{\text{LFMC},t+\tau} = h(x_{\text{forecast},t:t+\tau}; \psi)$$

where $y_{\text{LFMC},t+\tau}$ is the predicted LFMC at time $t+\tau$, $h(\cdot)$ is a time-aware machine learning model (e.g., temporal transformer, encoder-decoder LSTM, or Graph Neural Net), and X_{forecast} includes sequences of NWP or AI-NWP inputs such as:

$$X_{\text{forecast}} = \{\text{soil moisture}_\tau, \text{VPD}_\tau, \text{temperature}_\tau, \text{radiation}_\tau, \text{precipitation}_\tau, \text{humidity}_\tau, \dots\}$$

$T = 1, \dots, 10$ days.

Our approach builds on recent advances in high-resolution AI-NWP forecasting. For example, GraphCast (Lam et al., 2022) has demonstrated strong performance in multi-variable, global weather forecasting at sub-daily intervals, while FourCastNet (Pathak et al., 2022) offers efficient downscaling of weather dynamics with transformer-based architectures. These models have not yet been systematically applied to vegetation moisture dynamics, representing a key innovation in our proposal. By ingesting these forecasts into our trained LFMC models from Steps 1 and 2, we will create a predictive framework capable of generating fine-scale, future LFMC maps critical to fire behavior modeling. To ensure robustness and trust in these forecasts, we will: (i) Conduct **lead-time-stratified validation**, assessing performance at 3, 5, 7, and 10-day horizons. (ii) Compare NWP vs. AI-NWP performance and ensemble them when beneficial. (iii) Quantify **prediction uncertainty** using Monte Carlo dropout or ensemble variance techniques. We will explore two methodological paths: (i) **Direct Forecasting**: Using forecast variables as direct inputs to models trained to predict LFMC at future time points. (ii) **Recursive Forecasting**: Feeding one-step-ahead predictions iteratively to extend predictions over multiple time steps, with error accumulation monitored and constrained. This work is grounded in the growing literature linking climate forecasts to vegetation dynamics (e.g., Anderegg et al., 2020; Fang et al., 2021) and recent studies suggesting that LFMC can be predicted from lagged meteorological variables using machine learning (e.g., Joseph et al., 2023). However, those efforts have either relied on coarse-resolution inputs or ignored the short-term, high-resolution operational needs of fire agencies. Our proposed framework closes this gap by merging high-resolution environmental forecasts with site-specific LFMC prediction models, and producing **daily LFMC forecast maps at 10-meter resolution**—a capability not currently available in any operational or research-grade system.

Expected Outcomes of Step 3: (i) A validated system for 3–10 day LFMC forecasting at 10-meter spatial resolution. (ii) Comparative benchmarks of physics-based NWP vs. AI-NWP driven LFMC forecasts. (iii) Lead-time-specific performance profiles and uncertainty bounds. (iv) Forecast-driven dashboard layers supporting operational users in planning and response. By developing a forecasting pipeline that integrates modern environmental prediction with data-driven vegetation moisture modeling, this component transforms the system from a monitoring tool into a forward-operating, predictive decision support system—positioned to serve the real-time needs of CAL FIRE, USFS, and wildfire managers across the western U.S.

Step 4: Integration of Physics for Ecohydrological Realism: This stage enhances the predictive fidelity and generalization capacity of LFMC estimation by embedding physically grounded ecohydrological processes and states—such as evapotranspiration, atmospheric aridity, soil water availability, and plant water stress—into the machine learning modeling framework. Traditional data-driven approaches may fail to extrapolate reliably under anomalous climatic conditions or extreme events (e.g., droughts, heatwaves). To address this, we propose a hybrid modeling strategies that incorporate mechanistic understanding of the soil–plant–atmosphere continuum.

Key Inputs and Methods: We will obtain physically interpretable features through a soil water budget modeling pipeline that will simulate hydrologic fluxes (evapotranspiration, precipitation,

runoff, and deep percolation) and state variables (soil moisture) in the root zone. We will rely on state-of-the-art remote sensing-based evapotranspiration models (Melton et al., 2022) with diverse model physics and parameterizations, which is typically the largest and most uncertain hydrologic flux in the Western U.S. Appropriate soil properties will be obtained from public datasets (USDA) for soil water modeling. The output variables include root-zone soil moisture, plant-available water, and dynamic water deficit indices—key proxies for vegetation stress and fuel desiccation potential.

In addition, we will characterize historical episodes of extremely elevated atmospheric evaporative demand, referred to as “thirstwaves,” which represent a compound environmental condition capturing both radiative and aerodynamic components of the atmosphere (Kukul and Hobbins, 2025). These newly characterized events are hypothesized to often precede sharp declines in LFMC and have been underutilized in operational models. A specialized detection and characterization pipeline will be employed to identify thirstwave events and quantify their intensity, duration, frequency, and lagged impact on fuel moisture trends. These physics-based indicators are integrated into the LFMC modeling architecture in two ways: **(i) Feature Augmentation:** Ecohydrological outputs are added as covariates to machine learning models developed in Steps 1–3, improving their responsiveness to drought stress and atmospheric aridity. **(ii) Hybrid Modeling:** In select implementations, we embed physical constraints directly into the learning framework—e.g., by enforcing monotonic responses to water stress or bounding LFMC estimates based on modeled field capacity and wilting point thresholds.

Expected Benefits and Innovations: By coupling data-driven models with mechanistic process representations, this approach enhances model extrapolation under extreme or out-of-distribution scenarios, such as megadroughts or post-fire vegetation recovery. It also supports biophysical interpretability and builds stakeholder trust, especially among operational users seeking defensible predictions for resource allocation and risk communication. This integration marks a key advancement over purely statistical LFMC models by incorporating vegetation functional responses to hydrometeorological extremes. It also creates an adaptive modeling layer capable of informing tactical decision-making in variable and changing environments.

Expected Outcomes: (i) Incorporation of root-zone soil water budget variables and thirstwave indicators into LFMC prediction pipelines. (ii) Development of hybrid models with embedded physical constraints to improve realism under stress conditions. (iii) Improved forecast accuracy and robustness during drought periods or in novel climate regimes. (iv) Enhanced model interpretability and transparency for fire managers and stakeholders. This step grounds the broader AI- and SAR-driven system in biophysical reality, ensuring that model predictions reflect the actual moisture regulation processes in vegetation across ecosystems and climatic zones.

Step 5: Evaluation and Validation Plan: This section outlines our plan to evaluate and validate the proposed LFMC system, including how results will be analyzed and incorporated into model calibration and improvement. In this final stage of the project, we would emphasize rigorous field validation and calibration of the high-resolution Live Fuel Moisture Content (LFMC) estimation and forecasting system. While the remote sensing and machine learning pipeline developed in Steps 1 through 4 provides advanced, spatially explicit predictions of LFMC, these models must be empirically grounded to ensure accuracy, transferability, and stakeholder trust—particularly under extreme climatic conditions such as heatwaves and drought. To this end, we will implement a multi-site field validation strategy based on continuous ecohydrological monitoring using **LEAF (Logger for Environmental and Atmospheric Factors)** systems developed by Dr. Ashley Matheny’s lab (Matheny et al., 2019). The LEAF platform is a low-cost, modular sensor suite capable of continuously monitoring a variety of hydrometeorological and plant physiological variables relevant to LFMC dynamics. These include air temperature, precipitation, humidity, air pressure, **soil moisture**, **wood water content**, and **sap flow**, all measured at high temporal

resolution (typically every 30 minutes). These sensors can be solar- or battery-powered and stream data wirelessly to the cloud, supporting autonomous operation in remote field sites. LEAF systems have been successfully deployed in multiple ecosystems, including mangrove forests, oak savannas, and sites across the Texas Hill Country, demonstrating their robustness and scalability. For this project, we will deploy arrays of 10–20 LEAF systems at selected validation sites, focusing on areas with ecological importance and active wildfire management concerns, such as the **Balcones Canyonlands Preserve (BCP)** near Austin, Texas. These LEAF arrays will enable the continuous monitoring of LFCM and soil moisture (SM) in dominant vegetation species such as oaks and junipers, capturing the influence of forest structure, species composition, and microclimate on vegetation water status. Unlike traditional weekly destructive sampling, this sensor network will provide near-real-time tracking of diurnal and seasonal moisture dynamics, improving our ability to understand short-term fluctuations in LFCM that drive wildfire risk. We will complement these continuous observations with **periodic destructive sampling of LFCM**, collecting wet and dry biomass measurements to directly compare against model outputs and assess the accuracy of our machine learning predictions under varying moisture conditions. These sampling campaigns will be synchronized with **Sentinel-1 satellite overpasses** to ensure tight coupling between in-situ measurements and remote sensing-based LFCM estimates.



Figure 5: The LEAF-logger is a prototype sensor designed to measure tree water content at various depths developed by Dr Matheny's lab (Matheny et al 2019) which would be used for validation.

A key goal of this validation effort is to bridge the scale gap between point-level field data and satellite-based predictions at 10-meter resolution. We will conduct a detailed comparison between field observations and multiple remote sensing data products, including **NDVI and NDWI** from Sentinel-2 and Landsat-8/9, **soil moisture from SMAP**, and **vegetation evaporative stress from ECOSTRESS**. By evaluating how well these products track observed LFCM dynamics at the plot level and across heterogeneous landscapes, we aim to quantify the limitations and opportunities in using satellite proxies for LFCM at operational scales. Furthermore, this analysis will inform **scaling functions** and **correction models** that align point-scale LEAF measurements with pixel-scale satellite observations. The field datasets collected through this effort will be analyzed using standard statistical metrics, including **root mean square error (RMSE)**, **bias**, **coefficient of determination (R^2)**, and **power spectral density**. We will conduct **cross-validation across sites, species, and seasons**, with a particular emphasis on capturing the performance of the forecasting system under conditions of drought onset, peak fire season desiccation, and post-precipitation recovery. These results will guide the recalibration of machine learning and hybrid physics-AI models developed in earlier stages, ensuring that our predictions reflect realistic biophysical responses to environmental drivers.

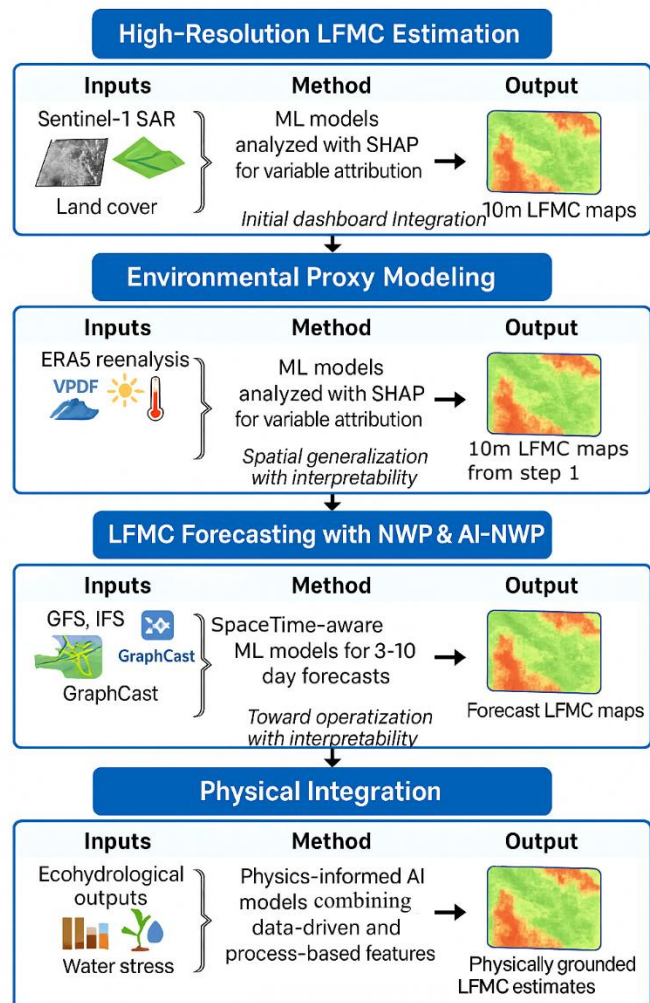
Importantly, this validation campaign builds upon Dr. Matheny's NSF CAREER Award and ongoing work in the BCP and broader Hill Country, which explores how LFCM and SM vary across spatial gradients, vegetation types, and ecohydrological settings. Her team's use of LEAF systems has already revealed critical insights into how tree species, mycorrhizal interactions, landscape position, and subsurface hydrology affect moisture retention and plant stress responses—factors highly relevant to wildfire vulnerability. Matheny's work also includes

integration of **leaf-, wood-, and soil-moisture data** to improve the remote detection and modeling of vegetation water status, with direct applications to the LFMC retrieval challenge addressed in this proposal. Moreover, the data generated through these LEAF deployments will be used not only for validation but also to improve satellite-based disaggregation of **vegetation optical depth (VOD)** and microwave soil moisture retrievals, contributing to the refinement of NASA and ESA's satellite products. These efforts will help disentangle the respective moisture contributions from leaves, wood, and soil—currently a major limitation in remote sensing of LFMC. By anchoring the proposed LFMC system in continuous, high-resolution field observations, this step ensures that model outputs are not only computationally advanced but also ecologically credible and operationally defensible.

4. Data Sources and Workflow

This project leverages a multi-source data architecture spanning active and passive remote sensing, environmental reanalysis, numerical forecasting, and in-situ observations. These datasets will support LFMC estimation, proxy modeling, forecasting, physical realism integration, and validation.

Data Type	Source	Use
Sentinel-1 SAR	ESA	VV and VH backscatter for high-resolution LFMC estimation
Landsat 8/9	USGS/NASA	Optical vegetation metrics (NDVI, NDWI) and surface temperature
ERA5 Reanalysis	ECMWF	Atmospheric forcing and soil/land state variables for proxy modeling
ECOSTRESS	NASA JPL	Evaporative stress index and transpiration proxies
SMAP (Soil Moisture Active Passive)	NASA	Root-zone and surface soil moisture for ecohydrological modeling
MODIS (MOD11, MOD13)	NASA	Moderate-resolution land surface temperature and vegetation products
GFS / IFS (NWP)	NOAA / ECMWF	Forecasted meteorological variables (VPD, RH, temperature, wind, etc.)
GraphCast, FourCastNet	Google DeepMind, NVIDIA	AI-driven short-term weather forecasts (1–10 day)
NOAA Analysis of Records Calibration (AORC)	NOAA	High-resolution meteorological datasets
USFS LFMC Field Observations	U.S. Forest Service	Destructive LFMC samples across CONUS for model training/validation
NLCD (National Land Cover Database)	USGS	Land cover classification for vegetation-specific modeling
USGS National Elevation Dataset	USGS	Elevation and slope inputs for terrain-based LFMC variation modeling
LEAF Sensor Networks	University of Texas at Austin , Matheny et al 2019	High-frequency measurements of xylem water content, sap flow, soil moisture
Thirstwave Index	Melton et al 2022	Event-based metrics capturing compound aridity/extreme evapotranspiration



To support the development of an operational LFM forecasting system, we leverage a multi-source data architecture and a modular AI pipeline structured across four core stages (Figure 5). These stages are designed to enable high spatial resolution, generalization beyond SAR availability, and operational forecasting capabilities, all grounded in physical realism. Stage 1: High-Resolution LFM Estimation from Sentinel-1: We begin by using Sentinel-1 synthetic aperture radar (SAR) data, combined with terrain and land cover inputs, to estimate LFM at 10-meter resolution. Machine learning models (e.g., XGBoost, TabNet, FT-Transformer) are trained using field-sampled LFM data and evaluated via spatial and temporal generalization metrics. These estimates will feed into an early version of the operational dashboard and serve as the baseline for subsequent modeling stages. Stage 2: Environmental Proxy Modeling (SAR-Free Estimation): In regions or periods where SAR data are unavailable, we develop proxy models using ERA5 reanalysis variables—including volumetric soil moisture, vapor pressure deficit (VPD), radiation, and temperature. These models are trained to estimate LFM at 10-meter

resolution, and analyzed using SHAP-based feature attribution to ensure interpretability and variable relevance. This approach expands the spatial and temporal applicability of the system, even under SAR limitations. Stage 3: LFM Forecasting Using NWP and AI-NWP: To extend beyond nowcasting, we incorporate numerical weather prediction (NWP) models (e.g., GFS, IFS) and AI-based forecasts (e.g., GraphCast, FourCastNet) as dynamic inputs. These models will enable 3–10 day forecasts of LFM at high spatial resolution, using time-aware ML architectures. We will also quantify lead-time-dependent uncertainty and validate against in-situ measurements and destructive LFM sampling data. Stage 4: Physical Integration: To embed physical realism and improve generalization under extreme or anomalous conditions, we integrate ecohydrological outputs such as soil water balance and plant water stress. These features will inform the design of physics-informed AI models that blend process-based and data-driven approaches. This hybridization supports better extrapolation in novel climatic or ecological contexts, enhancing stakeholder trust and operational utility. Operational Workflow: The overall system is designed for modular deployment, enabling LFM prediction via multiple pathways depending on data availability.

Figure 6: End-to-End Workflow for High-Resolution, Forecast-Enabled, and Physically Grounded LFM Estimation. This figure outlines the four-stage modeling pipeline developed in the project. The workflow begins with High-Resolution LFM Estimation using Sentinel-1 SAR and land cover inputs to train machine learning models for 10-meter LFM maps, with interpretability supported by SHAP analysis. In Environmental Proxy Modeling, ERA5 reanalysis variables serve as inputs to train ML models that generalize spatially in SAR-limited areas, using SAR-derived LFM as

training targets. The third stage, LPMC Forecasting, incorporates 3–10 day meteorological predictions from NWP and AI-NWP models (e.g., GFS, IFS, GraphCast) to generate forward-looking LPMC forecasts using spatiotemporal AI architectures. Finally, Physical Integration enhances realism by embedding ecohydrological features (e.g., water stress, soil moisture) into physics-informed ML models, yielding LPMC estimates that are both interpretable and operationally robust under extreme conditions.

The workflow begins with raw data ingestion and preprocessing, followed by ML/physics-based model execution, proxy or forecast substitution, field-based validation, and finally, delivery of output maps and decision-support layers through a real-time web dashboard for operational use.

5. Broader Impacts

Wildfires represent one of the most destructive and rapidly escalating natural hazards in the United States. As climate change continues to intensify fire behavior and shift fire regimes, public agencies and communities face increasing demand for timely, accurate, and localized information on vegetation moisture and fire risk. The proposed project contributes broadly to national wildfire resilience through actionable scientific innovations, workforce development, community engagement, and open-access technological infrastructure. **Deliverables** - This project will produce a set of tangible, high-impact deliverables that directly support fire science, management, and public safety operations: **(i) 10-meter resolution LPMC maps across CONUS**, calibrated and validated using ground observations, and physics-informed modeling. These maps will provide fire managers with a detailed and dynamic understanding of vegetation flammability at operational scales. **(ii) An operational LPMC forecasting system**, delivering daily nowcasts and 3–10 day forecasts of fuel moisture conditions. These predictions will support fire suppression planning, early warning systems, and resource prepositioning. **(iii) An open-source codebase and public APIs**, enabling reproducibility, transparency, and extensibility. Researchers, developers, and agency personnel will be able to integrate, adapt, and build upon the models, tools, and workflows created in this project. **(iv) A web-based dashboard** for real-time visualization and decision support. End users will be able to explore current and forecasted LPMC conditions interactively, filter by location or forecast horizon, and access context layers (e.g., land cover, recent fire perimeters) to inform operational decisions.

Extension of Existing Dashboard Capabilities: As part of this effort, we will build upon and significantly enhance the **existing NSF NCAR Fuel Moisture Content (FMC) Dashboard**, which currently provides VIIRS- and GOES-based LPMC retrievals at coarse resolutions (~375m). Our proposed dashboard extension will integrate: **(i) SAR-derived LPMC estimates at 10-meter resolution** for increased spatial precision; **(ii) Physics-informed machine learning forecasts**, extending lead times up to 10 days; **(iii) Ecohydrological drivers and uncertainty bounds**, enabling better interpretability and user trust; **(iv) Custom tools and layers** to support fire-specific workflows including ignition risk assessment, suppression planning, and public communication. These improvements will increase the value and usability of the dashboard for high-priority end users, including incident management teams, wildfire analysts, and municipal preparedness planners.

Stakeholder Engagement: Close engagement with fire, weather, and emergency management communities is critical to the success and uptake of the proposed system. To that end, we will establish collaborations with key operational and research partners including: **(i) CAL FIRE** (California Department of Forestry and Fire Protection), **(ii) USFS** (U.S. Forest Service), **(iii) NASA Applied Sciences Disaster Program**, **(iv) National Interagency Fire Center (NIFC)** and **(v) Wildland Fire Technology Innovation and Integration Center (WFTIIC)**. Through

these partnerships, we will ensure that system outputs align with real-world operational needs and that user feedback is directly incorporated into dashboard development and model refinement. Use cases include: (i) **Preposition Decision Support Tools**, which assist agencies in strategically staging firefighting assets; (ii) **Short-term fire behavior forecasting**, informed by localized LFM trends; (iii) **Public safety briefings**, based on forecasted vegetation desiccation patterns and ignition risk windows. We will convene **training workshops and user onboarding sessions**, providing hands-on exposure to the dashboard and modeling tools. These events will also serve as feedback loops, enabling us to gather input on interface design, forecast usability, and feature priorities. Additional engagement will occur through agency-led testbeds and seasonal wildfire outlook meetings.

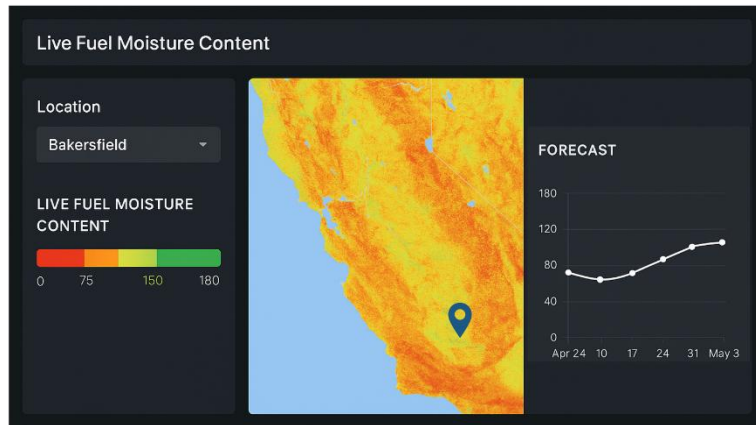


Figure 7: Conceptual interface of the proposed LFM forecasting dashboard, integrating 10-meter resolution nowcasts and 3–10 day forecasts derived from SAR, environmental proxies, and hybrid AI-physics models. The dashboard will support real-time visualization, uncertainty quantification, and decision-support layers tailored to operational fire management needs across the CONUS, with extensibility for regional applications

by different fire agencies.

Deliverables	Year 1				Year 2				Year 3			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Phase 1: System Kickoff and SAR-based LFM Estimation (Lead: Dr. Dashtian; Support: Dr. Singh, Postdocs)												
– Sentinel-1 ingestion + model training pipeline setup (Dashtian, Singh, Postdocs)												
– SAR-based LFM estimation model (LightGBM baseline) (Dashtian, Singh, Postdocs)												
Phase 2: Proxy Modeling for SAR-Limited Regions (Lead: Dr. Singh; Support: Dr. Kukal, Postdocs)												
– ERA5/AORC ingestion and feature engineering (Singh, Postdocs)												
– Proxy LFM model (SHAP-augmented) (Singh, Kukal, Postdocs)												
Phase 3: Forecasting with NWP and AI-NWP (Lead: Dr. Singh; Support: Dr. Dashtian, Postdocs)												
– Integration of GFS/IFS + GraphCast/FourCastNet (Singh, Postdocs)												
– Time-aware forecast model + uncertainty quantification (Singh, Dashtian, Postdocs)												
Phase 4: Physics Integration for Ecohydrological Realism (Lead: Dr. Kukal; Support: Dr. Matheny, Postdocs)												
– Root-zone modeling + thirstwave detection (Kukal, Matheny, Postdocs)												
– Hybrid AI-physics model development (Kukal, Singh, Postdocs)												
Phase 5: Validation, Documentation, and Release (Lead: Dr. Matheny; Support: Dashtian, Postdocs)												
– Full field validation (LEAF + destructive sampling) (Matheny, Postdocs)												
– Dashboard + open-source release + API documentation (Dashtian, Singh, Postdocs)												
Ongoing Reporting and Archiving (All PIs, led by Dashtian; Postdocs maintain logs)												
– Maintain GitHub logs, DOI minting, metadata tracking (All, led by Postdocs)												

6. Management Plan: This project will be led by Dr. Hassan Dashtian (PI, University of Texas at Austin), with key contributions from Dr. Manmeet Singh (Western Kentucky University), Dr.

Ashley Matheny (UT Austin), and Dr. Meetpal Kukal (University of Idaho), who bring complementary expertise in ecohydrology, machine learning, sensor development, and wildfire science. The team will follow a coordinated, modular development strategy organized around five technical phases aligned with the core components of the proposed system. Key roles and responsibilities are as follows: **Principal Investigator (PI): Dr. Hassan Dashtian:** Lead overall project coordination, reporting, and dashboard integration. Supervise AI modeling, forecasting, and environmental proxy components. Liaise with operational partners (e.g., CAL FIRE, USFS). **Co-Investigator: Dr. Ashley Matheny:** Lead LEAF sensor deployment, ecohydrological modeling, and field validation. Coordinate destructive sampling efforts and in-situ calibration activities. Ensure physiological relevance and sensor-based scaling alignment. **Co-Investigator: Dr. Manmeet Singh:** Oversee machine learning model implementation and optimization. Lead efforts on AI-NWP integration and temporal generalization. Contribute to forecast evaluation and uncertainty quantification. **Co-Investigator: Dr. Meetpal Kukal:** Integrate physics-based modeling, particularly thirstrwave metrics and hydrological budget. Support validation in western U.S. ecosystems and model hybridization. **Postdoctoral Researchers and Graduate Students:** Assist with data ingestion pipelines, model training, deployment, and fieldwork. Participate in technical development, dashboard design, and stakeholder engagement. We will conduct monthly team coordination meetings (virtual), quarterly technical progress reviews, and annual stakeholder workshops. Project management will be supported using GitHub (for code versioning and issue tracking), Slack (for rapid communication), and shared cloud repositories for data workflows. To ensure reproducibility and transparency, all modeling code, documentation, and key datasets will be publicly released via GitHub and registered via NASA Earthdata. The PI will submit all NSF-required reports on time and maintain open communication with the program officer regarding progress and risks.

7. Summary: This project aims to develop a next-generation, real-time Live Fuel Moisture Content (LFMC) forecasting system that combines synthetic aperture radar (SAR), environmental proxies, AI-driven weather forecasts, and ecohydrological modeling to produce daily 10-meter resolution nowcasts and 3–10 day forecasts across the continental U.S. Addressing critical gaps in spatial resolution, physical interpretability, and operational lead time, the system will integrate machine learning with physically grounded processes and be validated through extensive field measurements using LEAF sensor networks and destructive sampling. The resulting open-source, dashboard-ready platform will support fire agencies with actionable intelligence for ignition risk assessment, suppression planning, and public safety communication, transforming LFMC from a lagging variable into a predictive tool for wildfire management.

8. Results from Prior NSF Support

(i). RAPID: Hurricane Helene (2024)

Award #2502272 | PI: Hassan Dashtian, co-I Manmeet Singh | AGS | \$154,118 | Nov 2024–Oct 2025

Studied the hydrometeorological and societal impacts of Hurricane Helene, focusing on the Brown Ocean Effect and post-landfall storm intensification via wet soils. Integrated satellite data, AI-enhanced forecasts, and social media sentiment analysis. Outcomes include improved hybrid AI-physics models and insights into soil moisture's role in weather prediction, informing the proposed LFMC modeling.

(ii). CAREER: Biomass Hydraulic Capacitance & Transpiration

Award #2046768 | PI: Ashley Matheny | EAR | \$678,017 | Feb 2021–Jan 2026

Explores how trees store water to survive droughts using novel sensors and remote sensing. Developed LEAF systems, plant hydraulic models, and integrated data into drought/fire risk models. Supported public outreach via bilingual virtual field trips. Findings directly inform the proposed LFMC system with validated ground-truth data and modeling expertise.

(iii). I-Corps: Smart Sensing for Tree Water Stress

Award #2303014 | PI: Ashley Matheny | TI | \$50,000 | Feb 2023–Jul 2024

Commercialized a LEAF-based sensor for detecting tree water stress and wildfire risk. Conducted 110+ stakeholder interviews, refined a business model, and tested with local agencies. Demonstrated strong interest for use in wildfire readiness and ecosystem monitoring, enhancing real-world applicability of LFMC validation tools.

(iv). GEOPaths:GO – Earth Science Training in Jamaica

Award #2325619 | Co-PI: Ashley Matheny | ICER | \$529,977 | Oct 2023–Sep 2026

Develops an interdisciplinary, service-learning Earth science program in Jamaica. Dr. Matheny supports ecohydrology training and sensor integration. Outputs include new curricula, student-led virtual field trips, and hands-on conservation tech experience—advancing outreach and workforce development aligned with the LFMC system goals.